Unit FlashLight

SKU:U152



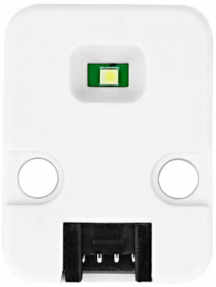


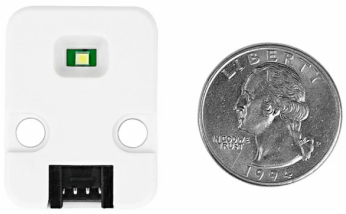














Description

Unit FlashLight is an output unit with a built-in flashlight, containing an AW3641 driver and a white LED with a color temperature of 5000 ~ 5700K. There is a mode selection switch on the internal PCB that allows you to choose between flash mode and constant light mode, using a GPIO input interface. It can be used as a flash source or for lighting applications.

Features

- PWM brightness adjustment
- Over-temperature alarm
- Overvoltage and short-circuit protection
- Two operating modes

Includes

- 1 x Unit FlashLight
- 1 x HY2.0-4P Grove connection cable (20cm)

Applications

- Light source
- DIY
- Alarm

Specifications

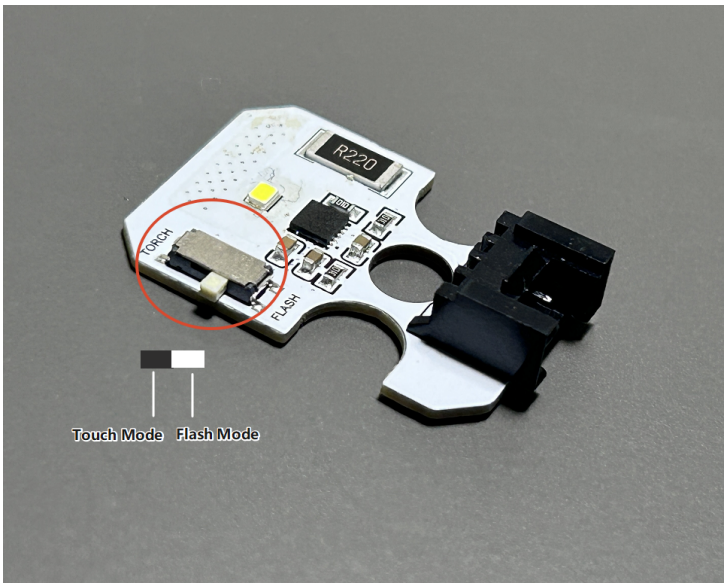
Specification	Parameter
Driver Chip	AW3641EDNR
Adjustment Method	PWM
AW3641EDNR Operating Temperature	-40℃ to +85℃
Unit Operating Temperature	0 ~ 40℃
Color Temperature	6000K~6500K (Kevin)
Product Size	32.0 x 24.0 x 8.0mm
Product Weight	4.6g
Package Size	138.0 x 93.0 x 9.0mm
Gross Weight	10.0g

Learn

Operating Mode Description

Unit Flashlight offers two operating modes. After removing the device enclosure, you can switch between modes using the DIP switch inside, following the silkscreen labels:

- Flash Mode: Pulse flash control for instantaneous high brightness. Supports setting 8 levels of flash intensity (100% ~ 30%) and flash timeout via single-wire protocol. Designed specifically for photo fill lighting.
- Torch Mode: Continuous steady illumination with PWM dimming support, allowing stepless brightness adjustment. Suitable for video recording or everyday lighting scenarios.

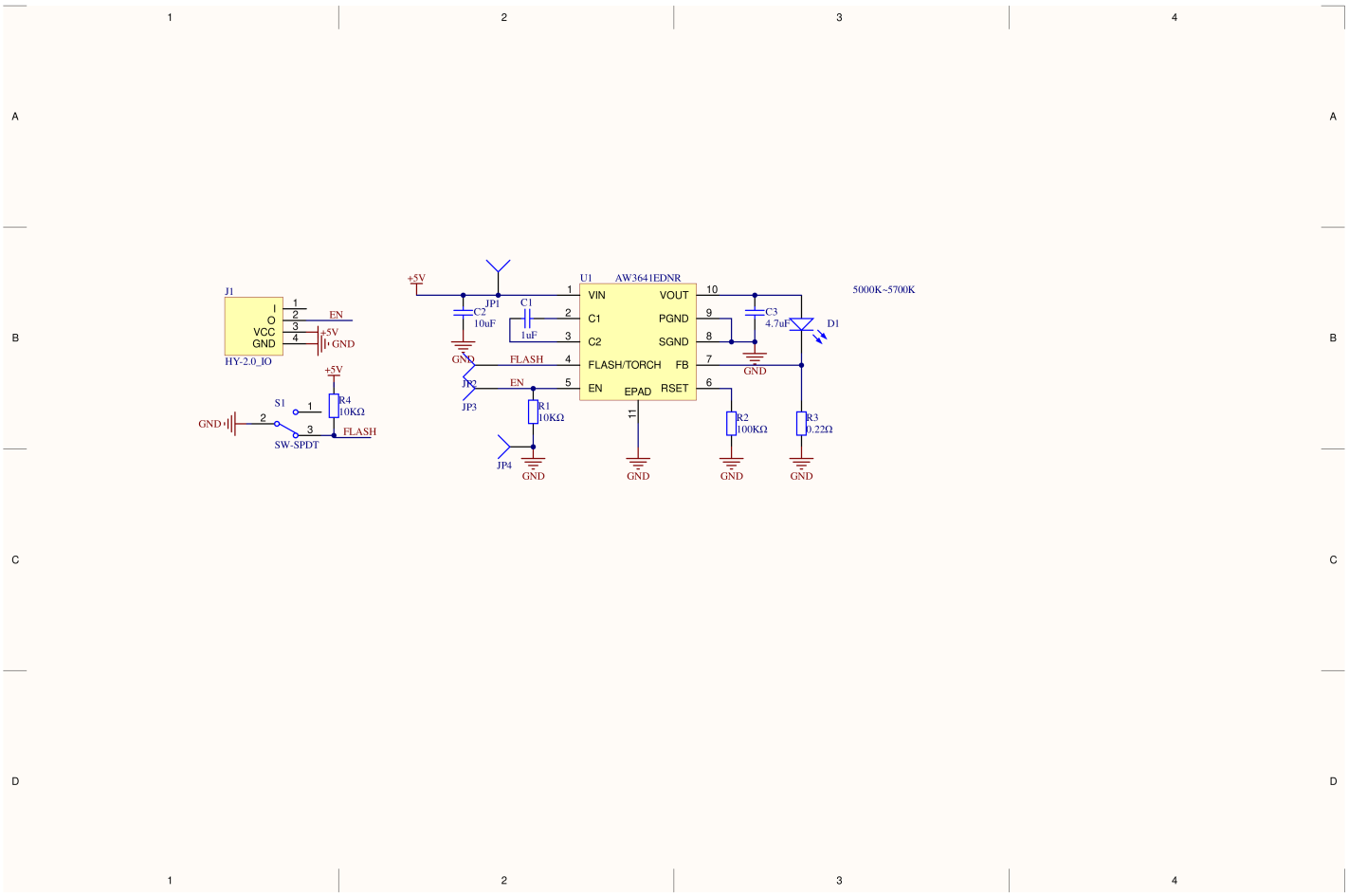


Note

In flash mode, the brightness is high and each flash generates heat. To prevent overheating and potential damage to the device, a interval of 8 ~ 15 seconds between flashes is recommended.

Schematics

- Unit FlashLight Schematics PDF

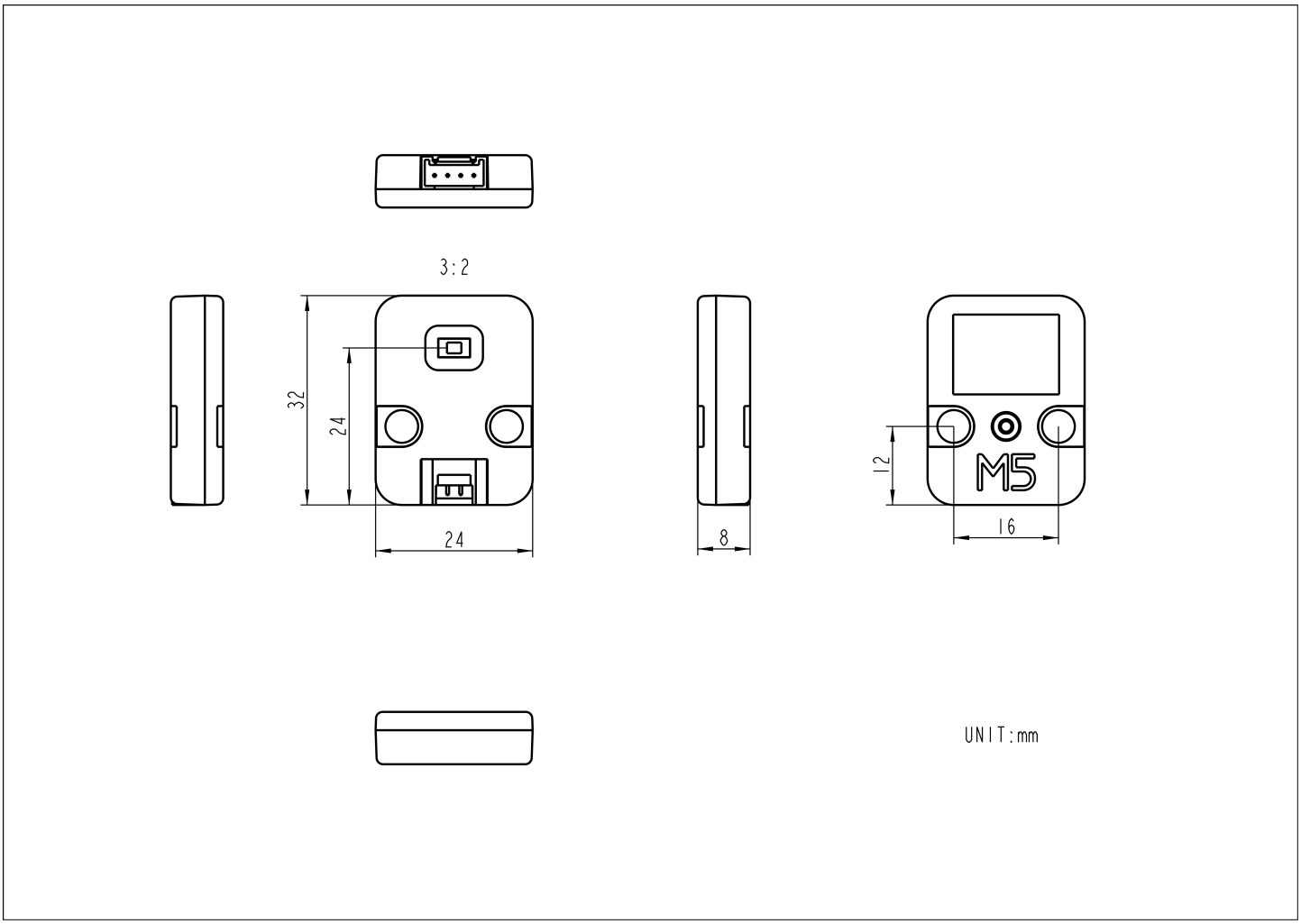


PinMap

Unit FlashLight

HY2.0-4P	Black	Red	Yellow	White
PORT.C	GND	5V	Light Control	NC

Model Size



Structure

- Unit FlashLight Structure Files

Datasheets

- AW3641EDNR Datasheet

Softwares

Arduino

- Unit FlashLight Test Example



UiFlow1

- Unit FlashLight UiFlow1 Docs

UiFlow2

- Unit FlashLight UiFlow2 Docs

Video